

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize in the Rocq prover two operational semantics for PROLOG with cut and prove them equivalent. The first semantics is based on a stack of alternative computations and is well suited for studying optimizations employed by PROLOG abstract machines; moreover, it closely models ELPI’s implementation. The second semantics is instead based on an and-or tree, in which the branches pruned by the cut operator are made explicit.

We use the tree-based semantics to reason about the determinacy analysis introduced in Elpi version 3.0, which statically identifies computations that produce at most one result.

2012 ACM Subject Classification Theory of computation → Operational semantics

Keywords and phrases Semantics, Prolog, Elpi, Determinacy Analysis, Rocq, Coq

Digital Object Identifier 10.4230/LIPIcs.CVIT.2016.23

Funding *Davide Fissore*: This work has been supported by the French government, through the France 2030 investment plan managed by the Agence Nationale de la Recherche, as part of the “UCA DS4H” project, reference ANR-17-EURE-0004.

1 Introduction

ELPI is a dialect of λ PROLOG (see [14, 15, 7, 12]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [13, 3, 4, 19, 8, 9]. Examples include the Hierarchy-Builder library-structuring tool [5] and Derive [17, 18, 11], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI gained a static analysis for determinacy [10] to help users tame backtracking. ROCQ users are familiar with functional programming but not necessarily with logic programming and uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checkers identifies predicates that behave like functions, i.e., predicates that commit to their first solution and leave no *choice points* (places where backtracking could resume).

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [10]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantic depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI’s implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [16] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations used by standard PROLOG abstract machines [20, 1], but it makes reasoning about the scope of cut difficult. To address that limitation we introduce a tree-based semantics in which the branches pruned by *cut* are explicit and we prove the two semantics equivalent. Using the tree-based semantics we then show that if every rule of a predicate passes the determinacy



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42nd Conference on Very Important Topics (CVIT 2016).

Editors: John Q. Open and Joan R. Access; Article No. 23; pp. 23:1–23:14

Leibniz International Proceedings in Informatics



LIPICs Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany

```

Inductive P := IP of nat. Inductive D := ID of nat. Inductive V := IV of nat.

Inductive Tm :=
  | Tm_P of P
  | Tm_D of D
  | Tm_V of V
  | Tm_App of Tm & Tm.

Record Unif := {
  unify : Tm → Tm → Σ → option Σ;
  matching : Tm → Tm → Σ → option Σ;
}.

```

■ Figure 1 Tm types

■ Figure 2 Unif type

44 analysis, then the call to the predicate does not leave any choice points.

45 2 Preliminaries: syntax and informal semantics

put unif and proe Before going to the two semantics, we show the piece of data structure that are shared
 gram in variable: by the them. The smallest unit of code that we can use in the language is an atom. The
 hides from types: atom inductive (see Type 1) is either a cut or a call. A call carries a term (see Figure 1). A
 term (Tm) is either a predicate, a datum, a variable or the binary application of two terms.

50 Inductive A := cut | call : Tm → A. (1)

51 Record R := mkR { head : Tm; premises : list A }. (2)

52 Record P := { rules : seq R; sig : sigT }. (3)

53 Definition Σ := {fmap V → Tm}. (4)

54 Definition bc : P → $\mathcal{F}_V$ → Tm → Σ → $\mathcal{F}_V$ * seq (Σ * seq A) := (5)

55 A rule (see Type 2) has a head of type term and a list of premises, that are a list of
 56 atoms. A program (see Type 3) is made by its rules and the signatures of predicates, **sigT**
 57 is a mapping from predicates to their signatures, signatures represent the definition of type
 58 from the simply typed lambda calculus, i.e. it is either a base type or an arrow for term
 59 application. We decorate arrows to know the mode (input or output) of the application
 60 argument.

61 A substitution (see Type 4) is a mapping from variables to terms. It is the output of a
 62 successful query. The empty substitution is noted ϵ .

63 The backchain function (bc, see Type 5) takes: a program P ; a set fv of variables ($\mathcal{F}_V$); a
 64 query q ; and the substitution σ . A backchain is performed when the current atom is a **call**.
 65 In this case, it start by recursively *dereferencing* the variables in σ , i.e. replacing all the
 66 variables in the query by their term mapping in σ .

67 TODO: free variables and program freshness

68 Then it takes the head of the term, which should be rigid, i.e. a predicate p , and look
 69 for the signature s of p in the program. Then, for each rule r in P , it unifies the head of r
 70 with q . Unification is possible thanks to the unificator U , whose type is shown in Figure 2.
 71 U explains how to unify terms up to standard unification (for output terms) or matching
 72 (for input terms). The function *unif* (resp. *matching*) unifies (resp. matches) the two terms
 73 under σ , it return an extension of σ if the two terms can be unified (resp. matched), and
 74 None otherwise. The result of a backchain is made by filtering the rules in P that can be used
 75 on the given query. In particular, for each rule filtered rules r , it returns a pair containing
 76 the extension of σ making the head of r and q unify and the list of premises of r .

```

f 1 2.    f 2 3.    r 0 0.    r 0 1.    r 0 2.
g X Z :- r X Y, f Y Z, !. % r1
g X Z :- f X Y, f Y Z.    % r2

```

Inductive tree :=
 | KO | OK | TA of A
 | Or of option tree & Σ & tree
 | And of tree & seq A & tree.

■ **Figure 3** Small ELPI program example

■ **Figure 4** Tree inductive

2.1 The cut operator

ELPI is a dialect of λ PROLOG with the hard cut operator and, in the language we have shown before, *cut* is an atom. The semantics of the cut operator adopted in the ELPI language corresponds to the *hard cut* operator of standard SWI-PROLOG. This operator has two primary purposes. First, it eliminates all alternatives that are created either simultaneously with, or after, the introduction of the cut into the execution state.

The ELPI program we will use as an example in the following sections is shown in Figure 3.

To illustrate the high-level description of *cut*, consider the program P shown in Figure 3 and the query $q = g \ 0 \ R$. Both rules for g can be used on the query q . Backchaining q in P from the empty (ϵ) substitution returns the list $[(\sigma_1, [r \ X \ Y, f \ Y \ Z, !]), (\sigma_2, [f \ X \ Y, f \ Y \ Z])]$ with σ_1 and σ_2 both equal to $\{X \mapsto 0, Z \mapsto R\}$. The two elements of the list are generated respectively from the rules r_1 and r_2 .

The next step of the interpretation corresponds to the backchaining of $r \ X \ Y$ from σ_1 . This returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [])]$ with $\sigma_3 := \sigma_1 + \{Y \mapsto 0\}$, $\sigma_4 := \sigma_1 + \{Y \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{Y \mapsto 2\}$. The three lists of premises are empty since all rules for r have no premises.

Backchaining $f \ Y \ Z$ in σ_3 gives no solution, therefore, by backtracking, we backchain $f \ Y \ Z$ in σ_4 . This gives a solution, namely $\sigma_6 := \sigma_4 + \{R \mapsto 2\}$.

We now reach the cut operator and can see its double side behavior. In the current execution state, we have still two unexplored alternatives: we can backtrack on the third solution for r that assign Y to 2 and on the rules r_2 . We can note, however, the first alternative have been generated after the cut introduction, and the second alternative have been generated at the same moment of the cut. Therefore both are cut away leaving no choice point. The final complete solution is: $\{X \mapsto 0, Z \mapsto R, Y \mapsto 1, R \mapsto 2\}$, where X , Y and Z can be ignored since they are the result internal unification.

We propose two formal, operational semantics for a logic program with cut. The two semantics are based on different syntaxes, the first syntax (called *tree*) exploits a tree-like structure and is ideal both to have a graphical view of its evolution while the tree is being interpreted and to prove lemmas over it. The second syntax, called *elpi*, is the ELPI's syntax and has the advantage of reducing the computational cost of cutting and backtracking alternatives by using shared pointers. We aim to prove the equivalence of the two semantics together with some interesting lemmas of the cut behavior.

Notations used in the paper In the following section we note $\mathbb{B}$ for the boolean type, and $\top$ and $\perp$ for **true** and **false**. The option type is noted $T + 1$ where T is the type carried by the option instance, $\boxed{t}$ corresponds to *Some t* and $\square$ corresponds to *None*.

3 And-Or Tree semantics

The tree state The tree inductive, shown in Figure 4. We distinguish 5 main cases: KO , OK , and are special meta-symbols representing, respectively, the basic failed and a successful

se metti $r1 = g \ A \ B :- f \ A \ B$. allora $g \ e \ f$ sono funzioni, e puoi spiegare anche l'idea del detcheck qui

```

Fixpoint get_end s A :  $\Sigma$  * tree :=
  match A with
  | TA _ | KO | OK  $\Rightarrow$  (s, A)
  | Or  $\square$  s1 B  $\Rightarrow$  get_end s1 B
  | Or (Some A) _ _  $\Rightarrow$  get_end s A
  | And A _ B  $\Rightarrow$ 
    let (s', pA) := get_end s A in
    if pA == OK then get_end s' B
    else (s', pA)
  end.

Definition get_subst s A := (get_end s A).1.
Definition path_end A := (get_end  $\epsilon$  A).2.
Definition success A := path_end A == OK.
Definition failed A := path_end A == KO.
Definition path_atom A :=
  if path_end A is TA _ then  $\top$ 
  else  $\perp$ .

```

(a) Definition of *get_end*(b) Functions from *get_end*

■ **Figure 5** *get_end* and derives functions

115 trees. These symbols are considered meta because they are internal intermediate symbols
 116 used to give structure to the tree.

117 The *TA* constructor (acronym for tree-atom) is the constructor of atoms in the tree. They
 118 are either the cut operator or a call, in these cases, the interpreter should evaluate the atom
 119 by either cutting the subtrees in the scope of the cut or backchaining the call in the program.

120 The two recursive cases of a tree are the *Or* and *And* nodes. The *Or* node $A \vee B_\sigma$ denotes
 121 a disjunction between two trees *A* and *B*. The first branch is optional, if absent it represents
 122 a dead tree, i.e. a tree that has been entirely explored. The second branch is annotated
 123 with a suspended substitution σ so that, upon backtracking to *B*, σ is used as the initial
 124 substitution for the execution of *B*.

125 The *And* node $A \wedge_{B_0} B$ represents a conjunction of two trees *A* and *B*. We call B_0 the
 126 reset point for *B*; it is used to restore the state of *B* to its initial form if a backtracking
 127 operation occurs on *A*. An example of this behavior is shown in Section 3.2

128 A graphical representation of a tree is shown in Figure 6a. To make the graph more
 129 compact, the *And* and *Or* nodes are n-ary rather than binary, and the children of these
 130 nodes associate to the right. The *KO* node acts as the neutral elements in the *Or* list, while
 131 *OK* is the neutral element of the *And* list.

132 *The tree interpreter* The interpretation of a tree is performed by two main routines: *step*
 133 and *next_alt* that traverse the tree depth-first, left-to-right. Then, then *runT* inductive
 134 makes the transitive closure of *step* and *next_alt*: it iterates the calls to its these
 135 auxiliary functions as much as needed. In Types 7–9 we give the types contrats of these
 136 symbols where $\mathcal{F}_v$ is a set of variable names.

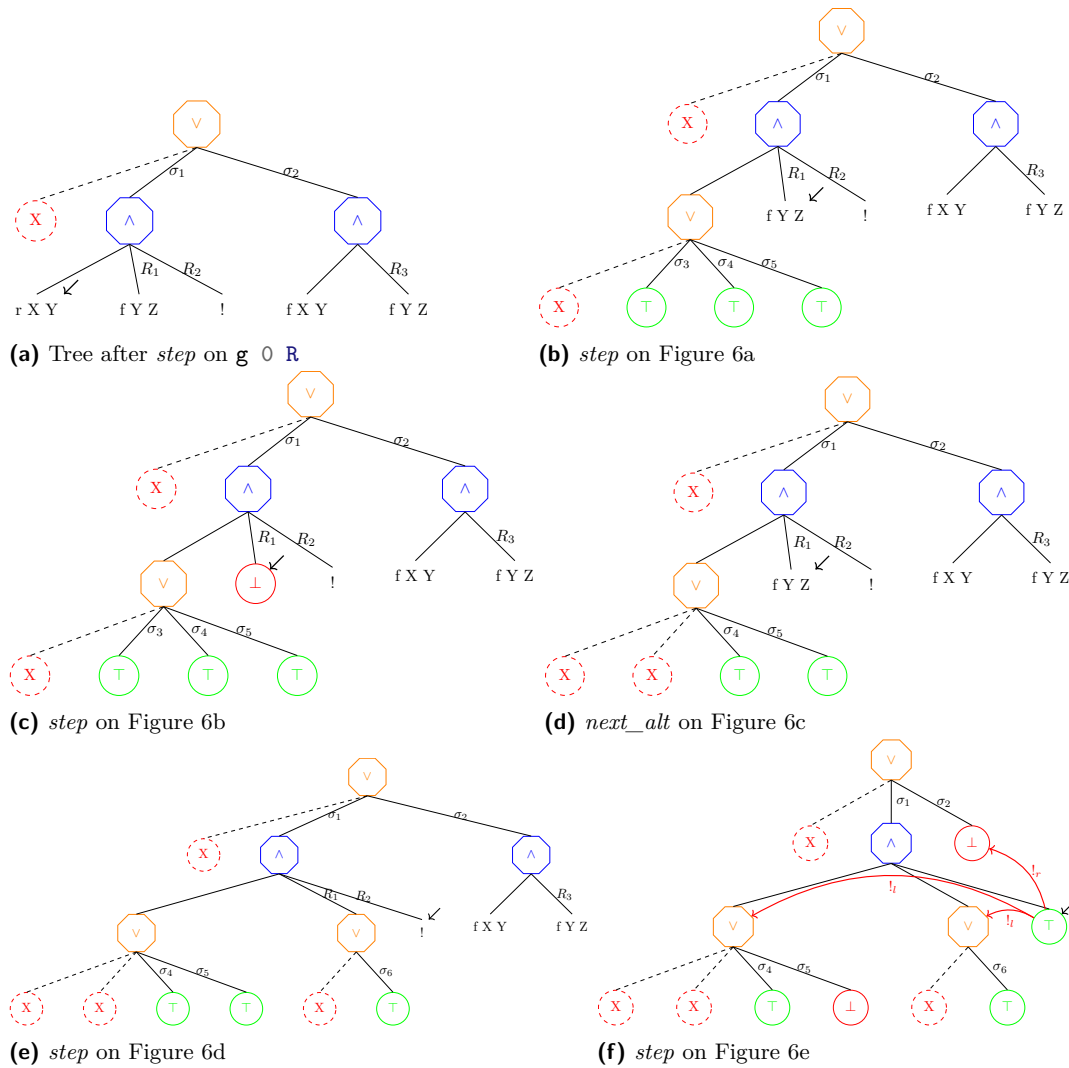
137 **Inductive** tag := Expanded | CutBrothers | Failed | Success. (6)

138 **Definition** step : $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F}_v * \text{tag} * \text{tree}) :=$ (7)

139 **Definition** next_alt : $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=$ (8)

140 **Inductive** runT (p : $\mathbb{P}$): $\mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree}$
 $\rightarrow \text{option } (\Sigma * \text{option tree}) \rightarrow \mathbb{B} \rightarrow \mathcal{F}_v \rightarrow \text{Prop} :=$ (9)

141 The tree interpreter, as in prolog, explores the tree in DFS strategy, to discover the
 142 substitution and the leaf of the tree that should be interpreted. These two pieces of
 143 information can be retrieved for a substitution *s* and a tree *A* thanks to the *get_end* routine,
 144 shown in Figure 5a If the tree is already a leaf, *get_end* returns its inputs. Otherwise, if the
 145 tree is a disjunction, the path continues on the left subtree, if it exists, otherwise *get_end*
 146 contues on the ths using the substitution stored in the *Or* node. In the case of a conjunction,



■ **Figure 6** Some tree representations¹

147 if *get_end* of the lhs returns the tree *OK*, then the *get_end* continues in the rhs, otherwise
 148 we return the result of the lhs. We derive some useful definition from *get_end*. They are
 149 listed in Figure 5b.

150 We have pointed by a black arrow the *path_end* of all the trees in Figure 6.

151 3.1 The *step* procedure

152 The *step* procedure takes as input a program, a set of variables, a substitution, and a tree,
 153 and returns an updated set of variables, a *tag*, and an updated tree.

154 The set of variables v given in input to *step* are used and updated when a backchaining
 155 operation takes place. A call to *step* produces a good result if v contains all the variables

¹ The substitution $\sigma_1 \dots \sigma_6$ are from Section 2.1, R_1, R_2 and R_3 are the reset points and correspond respectively to the list of atoms $[f \ Y \ Z, !], [!]$ and $[f \ Y \ Z]$

not true in fig₇ in the tree. This guarantees that any call to *bc* returns a list of atoms whose variables are disjoint from the variables of the current working tree.

ure:fix + also fix₈ The *tag* (see Type 6) indicates the kind of an internal tree step: **CutBrothers** denotes the interpretation of a superficial cut, i.e., a cut whose parent nodes are all *And*-nodes. **Expanded** denotes the interpretation of non-superficial cuts or predicate calls. **Failure** and **Success** are returned for, respectively, *failed* and *success* trees.

162 The step procedure is intended to evaluate atoms, that is, it leaves unchanged the tree iff its *path_end* is not an atom, i.e. if it is a success- or a failed-tree.

164 **Lemma** *succ_step_iff*: $\forall p \ v \ s \ A, \text{success } A \leftrightarrow \text{step } p \ v \ s \ A = (v, \text{Success}, A).$ (1)

165 **Lemma** *fail_step_iff*: $\forall p \ v \ s \ A, \text{failed } A \leftrightarrow \text{step } p \ v \ s \ A = (v, \text{Failed}, A).$ (2)

166 The interpretation of a call to a term *c* starts by calling the *bc* function on *c*. The result of the backchain is a list *l*. If *l* is empty then the current call is replaced by the *KO* node, otherwise it is replaced by a right skewed tree made by *Or* inner-nodes and each leave correspond to a list of atom *e* in *l*. If *e* is empty then it is mapped to the *OK* tree, otherwise it is mapped to a right-skewed tree made by *And* inner-nodes.

sottolineare le In Figure 6a, we provide a clear example of a call to *step* on the query *q 0 R* on the sostituzioni 172 program *P* in Figure 3. Let *c*₀, *c*₁ and *c*₂ be the children of the root. *c*₀, by construction is 173 the *None* tree, while *c*₁ and *c*₂ correspond respectively to the premises of rules *r*₁ and *r*₂ in 174 *P*. We need the first *None* subtree so that we can attach a substitution to *c*₁. In particular, 175 we note that the *c*₁ (resp. *c*₂) are related to the substitution σ_1 (resp. σ_2). The value of σ_1 176 and σ_2 are the same as in Section 2.1. The reset points in the tree are marked with the *R* 177 symbol. $R_1 := [\mathbf{f} \ Y \ Z, !]$, $R_2 := [!]$ and $R_3 := \mathbf{f} \ Y \ Z$, they are essentially the list of atoms 178 that allowed to generate the current subtree. Other example of call to the *step* procedure 179 triggering a backchain are Figures 6b, 6c, and 6e.

180 The interpretation of a cut has three main impacts: at first the cut is replaced by the 181 *OK* node, then some special subtrees, in the scope of the *Cut*, are cut away: in particular we 182 need to soft-kill the left-siblings of the *Cut* and hard-kill the right-uncles of the the *Cut*.

183 ► **Definition 1** (Left-siblings (resp. right-sibling) and Right-uncles). Given a node *A*, the 184 left-siblings (resp. right-sibling) of *A* are the list of subtrees sharing the same parent of *A* 185 and that appear on its left (resp. right). The right-uncles of *A* are the list of right-sibling of 186 the father of *A*.

187 ► **Definition 2** (Hard-kill, $!_r$ and Soft-kill, $!_l$). Given a tree *t*, hard-kill replaces the given 188 subtree with the *KO* node. If *t* is a success-tree, then soft-kill replaces with *KO* all subtrees 189 that are not part of the path in *t* leading to the *OK* node.

190 An example of the impact of the cut is show in Figure 6f. In Figure 6e, we see that 191 the *path_end* is a cut. The cut is replaced by the *OK* node and it need to cut-away some 192 alternatives in the tree. The red arrows indicate the scope of the cut, they are labeled with 193 $!_r$ and $!_l$ to explain which kind of cut has been performed on each pointed subtree. We note 194 that, after the evaluation of a cut, the path leading to the cut node is not affected by $!_l$, 195 the success under the substitution σ_4 is kept, while the success under the substitution σ_5 is 196 replaced by the *KO* node. If we refer to the paragraph in Section 2.1, we are discarding the 197 third rule of the *r* predicate.

198 3.2 The *next_alt* procedure

199 The *step* procedure alone is not sufficient to reproduce entirely the behavior of the full 200 expected prolog interpreter. As proved in Lemma 1 and Lemma 2, *step* does not make any

progression on the tree being interpreted. To solve this problem, we need a routine capable of backtracking in case of failures. This routine return the backtracked tree if it exists, *None* otherwise.

In Figure 6c, we are facing a failure: the interpreter has evaluated the atom $\mathbf{f} \ X \ Y$ under the substitution σ_3 which assigns Y to 0, but, in Figure 3, we have no rule satisfying this query. The role of *next_alt*, in this example, is to kill the path leading to this failure, while resetting the failure to its original shape. We see how the reset point plays a major role in this case. At first we are killing the *OK* node under the substitution σ_3 , since this tree have been entirely explored and led to no solution, then we replace the *KO* node with the information stored in R_1 . That is, thanks to R_1 , we are restoring the $\mathbf{f} \ Y \ Z$ call, so that we are able to evaluate this node under the substitution σ_4 . More concretely, we have uncessesfully tried rule $\mathbf{r} \ 0 \ 0$, therefore, we need to continue by trying rule $\mathbf{r} \ 0 \ 1$.

Furthermore, *next_alt* accomplishes to another important task. Its behavior changes depending to the value of the boolean it receives in entry: *next_alt false A* aims to recursively cancel all the failures (if any) in A , whereas, *next_alt true A* aims to cancel the first success in A . This second behavior is helpful since, we need to progress the tree in case of success. For example, consider the tree in Figure 6f, it contains a success, mapping namely $\mathbf{R}$ to 2. The interpreter should return this success, but, due to the “non-deterministic” behavior of the prolog system, we also need to return all the other solution that could be available by evaluating the rest of the tree. After the replacement of the *path_end* in Figure 6f with *KO*, we remain with a tree having all failing path. The *next_alt* procedure keeps removing them until discovering that the tree have no worth-to explore paths, therefore it will return *None*.

Some interesting property of *next_alt* are shown below and allow to see how *next_alt* complements *step* wrt failed, success and *path_atom* trees.

Lemma *path_atom_next_alt_id* $\mathbf{b} \ A$: $\text{path_atom } A \rightarrow \text{next_alt } \mathbf{b} \ A = \text{Some } A$. (3)

Lemma *next_alt_failedF* $\mathbf{b} \ A \ A'$: $\text{next_alt } \mathbf{b} \ A = \text{Some } A' \rightarrow \text{failed } A' = \perp$. (4)

Lemma *next_altFN_fail*: $\forall A, \text{next_alt } \perp \ A = \square \rightarrow \text{failed } A$. (5)

3.3 The *runT* inductive

The inductive *runT* is modeled as a function taking 4 inputs and returning 3 outputs. The inputs are the program, a set of variables v_0 , an initial substitution σ_0 , and a tree t_0 . The outputs are an optional object r containing a substitution σ_1 and an optional tree t_1 , and a boolean b and a set of updated variables v_1 . The very interesting piece of information is the result r , if *None*, it tells that the interpretation failed, otherwise, σ_1 represents the most-general unificator that makes the execution of the tree t_0 under the initial substitution σ_0 succeeds: σ_1 is an extension of σ_0 . The tree t_1 is the updated tree containing the alternatives that have not yet been explored, if these alternatives do not exists, t_1 is *None*. The output b and v_1 are there to tell respectively if during the execution a superficial cut has been evaluated, and what is the updated set of variables after the interpretation.

The procedure *runT* is based on four main derivation rules, shown in Figure 7. (STOPT) If the *path_end* of the tree t_0 is a success, the substitution σ_1 is retrived in t_0 from σ_0 and the input tree is cleaned of its successful path. (STEPT) If the *path_end* of t_0 is an atom, then *step* is invoked to evaluate t_0 , and *runT* is recursively called on the new tree. (BACKT) If the *path_end* of t_0 is a failure, *next_alt* is called to clear the failed path; if the resulting cleaned tree t_1 exists, *runT* is recursively called on it. Finally, (FAILT) takes into account tree with no solutions.

replace *None*
with $\perp$

246 **Lemma** `runT_det`: $\forall p \ v_0 \ s_0 \ t_0 \ r_1 \ r_2 \ b_1 \ b_2 \ v_1 \ v_2,$
`runT p v_0 s_0 t_0 r_1 b_1 v_1` $\rightarrow$ `runT p v_0 s_0 t_0 r_2 b_2 v_2  $\rightarrow r_2 = r_1 \wedge v_2 = v_1 \wedge b_2 = b_1$ . (6)`

247 The lemma above tells that `runT` behaves deterministically. The proof is by induction
 248 on the first `runT` execution and the proof is rather immediate, since all rules are in exclusive
 249 due to the hypothesis on the `path_end`: the hypotheses `success t_0`, `path_atom t_0` and `failed t_0`
 250 are exclusive. Moreover, by Lemma 6, we have that the hypothesis of `FAILT` implies `failed`
 251 `t_0`, therefore it is exclusive wrt `STOPT` and `STEPT`, but also from `BACKT` since they have
 252 different expectations on the output of `next_alt`.

$$\begin{array}{c}
 \frac{\text{success } t_0 \quad \text{get_subst } s_0 \ t_0 = s_1 \quad \text{next_alt } \top \ t_0 = t_1}{\text{runT } v_0 \ s_0 \ t_0 \ (\text{Some } (s_1, t_1)) \perp v_0} \text{STOPT} \\
 \\
 \frac{\text{path_atom } t_0 \quad b_2 = \text{is_cb } \text{tg} \parallel b_1 \quad \text{step } p \ v_0 \ s_0 \ t_0 = (v_1, \text{tg}, t_1) \quad \text{runT } v_1 \ s_0 \ t_1 \ r \ b_1 \ v_2}{\text{runT } v_0 \ s_0 \ t_0 \ r \ b_2 \ v_2} \text{STEPT} \\
 \\
 \frac{\text{failed } t_0 \quad \text{next_alt } \perp \ t_0 = \text{Some } t_1 \quad \text{runT } v_0 \ s_0 \ t_1 \ r \ n \ v_1}{\text{runT } v_0 \ s_0 \ t_0 \ r \ n \ v_1} \text{BACKT} \\
 \\
 \frac{\text{next_alt } \perp \ t_0 = \square}{\text{runT } v_0 \ s_0 \ t_0 \ \square \perp v_0} \text{FAILT}
 \end{array}$$

■ **Figure 7** Rule system for `runT`

253 3.4 Properties of `runT`

254 Working with `runT` allows to prove some interesting properties about the behavior of the
 255 cut operator wrt disjunction, i.e. the `Or` node. We have proven several properties, we cite
 here some that are quite interesting. They are available *here*.

howkward... use
 another style

256 **Lemma** `runSST_or`: $\forall p \ v_0 \ v_1 \ s_0 \ s_1 \ t_0 \ t_0' \ \text{sm } t_1,$
`runT p v_0 s_0 t_0 (Some (s_1, Some t_0'))` $\top v_1 \rightarrow$
 257 `let sR := Some (Or (Some t_0') sm KO) in`
`runT p v_0 s_0 (Or (Some t_0) sm t_1) (Some (s_1, sR))` $\perp v_1$. (7)

Lemma `run_orSST` `p v_0 v_2 s_0 s_1 sm t_0 t_0' t_1 t_1'`:
 258 `runT p v_0 s_0 (Or (Some t_0) sm t_1) (Some (s_1, (Some (Or (Some t_0') sm t_1'))))` $\perp v_2 \rightarrow$
 $(\exists b, \text{runT } p \ v_0 \ s_0 \ t_0 \ (\text{Some } (s_1, \text{Some } t_0')) \ b \ v_2 \wedge t_1' = \text{if } b \text{ then } KO \text{ else } t_1).$ (8)

259 ► **Lemma 3** (`runSST_or`). *xx*

260 In the first lemma we claim that if we run a tree `t_0`, and the interpretation returns a
 261 success after executing a superficial cut (note the `true` boolean), the interpretation of the
 262 tree `t_0 t_1` still return a success, with the same σ_1 , the output tree still exists, but the rhs has
 263 been cut-away due to the execution of the superficial cut in `t_0`.

264 Conversely, if the term `t_0` $\vee$ `t_1` evaluates to a substitution σ_0 and know that `t_0` leaves
 265 alternatives (namely `t_0'`), then we know that evaluating `t_0` alone will produce the same
 266 substitution σ_0 and alternatives `t_0'`, but, depending on if the evaluation of `t_0` has crossed a
 superficial cut, we know if the state `t_1` will be cut-away or not in the evaluation of `t_0` $\vee$ `t_1`.

dire che se non ci
 sono cut super-
 ficiali allora si
 comporta come
 or della logica
 normale.

Definition $\text{stepE } v_0 \text{ t s a gl} :=$
 $\text{let } (v_1, \text{rs}) := \text{bc p } v_0 \text{ t s in}$
 $(v_1, \text{save_alts a gl (r2a rs)}).$

Inductive $\text{runE} : \mathcal{F}_v \rightarrow \text{alts} \rightarrow \text{option } (\Sigma * \text{alts}) \rightarrow \text{Prop} :=$ (10)

$$\begin{array}{c}
\frac{}{\text{runE } v_0 ((s_0, [::]) :: a) (\text{Some } (s_0, a))} \text{STOPE} \\
\\
\frac{\text{runE } v_0 ((s_0, \text{gl}) :: ca) r}{\text{runE } v_0 ((s_0, [:: (\text{cut}, ca) \& \text{gl}]) :: a) r} \text{CUTE} \\
\\
\frac{\text{stepE } v_0 \text{ t } s_0 \text{ al gl} = (v_1, [:: b \& \text{bs}]) \quad \text{runE } v_1 (b :: \text{bs}++\text{al}) r}{\text{runE } v_0 ((s_0, [:: (\text{call } t, ca) \& \text{gl}]) :: al) r} \text{CALLE} \\
\\
\frac{\text{stepE } v_0 \text{ t } s_0 \text{ al gl} = (v_1, [::]) \quad \text{runE } v_1 \text{ al } r}{\text{runE } v_0 ((s_0, [:: (\text{call } t, ca) \& \text{gl}]) :: al) r} \text{BACKE} \\
\\
\frac{}{\text{runE } v_0 [::] \square} \text{FAILE}
\end{array}$$

■ **Figure 8** Rule system for runE

4 Stack semantics

We now introduce the ELPI semantics. The interpreter is close to that of the ELPI language and provides an operational semantics that is similar to the one proposed by Pusch in [16], which is in turn closely related to the semantics given by Debray and Mishra in [6, Section 4.3]. Pusch mechanized this semantics in Isabelle/HOL, together with several optimizations that are also present in the Warren Abstract Machine [20, 1].

We represent a state of the ELPI language thanks to the two mutual inductives shown below.

Inductive $\text{alts} := \text{no_alt} \mid \text{more_alt of } (\Sigma * \text{goals}) \& \text{alts}$
with $\text{goals} := \text{no_goals} \mid \text{more_goals of } (A * \text{alts}) \& \text{goals}.$

An ELPI state is an enhanced two-dimensional list. The outermost list represents a list of alternatives in disjunction, each accompanied by the substitution that should be used for their interpretation. The innermost list is a list of atoms, that is, the list of goals in conjunction. These goals are decorated with a pointer to an alternative list, which is used to keep track of where to jump when a cut is interpreted. We call these special alternatives the *cut-to* alternatives.

In our implementation of

The idea of the ELPI interpreter is to receive a list of alternatives. The first alternative consists of a list of goals. Four cases must be taken into account; they are shown in Figure 8. In order to simplify goal retrieval, we split the head of the alternatives from the tail, so that it can be immediately matched in the inductive definition. Note that an empty list of alternatives represents, by definition, a failing elpi state. If the goal list is empty (STOPE), then we have, by definition, a success, and the input solution together with the list of alternatives is returned. If the goal list starts with a cut (CUTE), then the current

alternatives are erased in favour of the cut alternatives, and a recursive call is made on the remaining goal list.

Finally, we must consider the case in which the goal list starts with a call. The call can either fail (FAILE) or succeed (CALLE). We distinguish the two cases by looking if the backchaining operation returns zero or more rules. We have wrapped this task in the *stepE* procedure, which also updates the goal and cut-alternative list. The fail case, is relatively easy: the first goal does not succeed, we need to take the head of the alternatives, and make it the new list of goals to be explored.

The case in which backchaining produces a non empty list, the *save_alts* routine is in charge of: taking the list of premises and add to each atom the the list of alternatives *a* as their new cut-alternatives, then it append the list of goals *gl* to each of these new lists.

4.1 XXX

It is not possible to prove the same lemma as in section xxx, because no structure, the scope of the cut ... spiegare bene il problema so that i can reuse this subsection has a hook when I will say that l2t is difficult + Yves

5 Equivalence of the two semantics

The equivalence between the two semantics is possible under two conditions: we need to work with “valid tree”, i.e. all of those trees that can be generated from a call. Secondly, we need to translate trees into elpi states. In the next to subsection we propose the two functions *valid_state* and *t2l*.

5.1 Valid trees

The inductive tree allows one to generate a large number of trees, some of which are not valid, in the sense that they cannot be produced starting from a given query. The class of valid trees is characterized by the function shown in Figure 9a.

While all base cases of tree are considered valid, we need to analyze carefully the cases for the *Or* and *And* constructors.

For the *Or* constructor, we distinguish two cases depending on whether the left-hand side (lhs) exists. If it does not exist, then the right-hand side (rhs) must be a valid tree. Otherwise, the lhs must itself be a valid tree, and the rhs is either the *KO* tree, since it may have been removed by the evaluation of a superficial cut in the lhs, or it has not yet been explored. In the latter case, it is a **base_or** tree, namely the right-skewed tree formed by a disjunction of conjunctions that is generated after a successful backchain to a call.

For the *And* constructor, the lhs is required to be a valid tree. The shape of the rhs depends on whether the lhs is a success. If the lhs is not successful, then the rhs has never been explored: the procedures *step* and *next_alt* modify the rhs only when the lhs succeeds. In this case, the lhs must be the right-skewed tree containing conjunctions of premises atoms, the reset point B_0 ensure what shape the rhs should have. If the lhs is a success tree, then the rhs can be modified by *step* and *next_alt*, therefore it must be a valid tree.

5.2 From trees to lists

The translation of a tree to a list is shown in Figure 9b. It takes the tree to be translated, a substitution (called *s*), a list of alternatives, called *bt*. The substitution *s* tells what is the

```

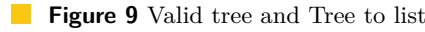
Fixpoint t2l A s0 bt :=
  match A with
  | OK          => [:: (s0, [::])]
  | KO          => [::]
  | TA a        => [:: (s0, [:: (a, [::])])]
  | Or □ s1 B => add_ca_deep bt (t2l B s1 [::])
  | Or (Some A) s1 B =>
    let lB := t2l B s1 [::] in
    let lA := t2l A s0 lB in
    add_ca_deep bt (lA ++ lB)
  | And A B0 B =>
    let lA := t2l A s0 bt in
    let lB0 := a2g B0 in
    let lA := add_deep bt lB0 lA in
    if lA is [:: (s0, gs) & al] then
      let al := map (catr lB0) al in
      let lB := t2l B s0 (al ++ bt) in
      map (catl gs) lB ++ al
    else [::]
  end.

Fixpoint valid_tree A :=
  match A with
  | TA _ | OK | KO => ⊤
  | Or □ _ B => valid_tree B
  | Or (Some A) _ B => valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A B0 B => valid_tree A &&
    if success A then valid_tree B
    else B == big_and B0
  end.

```

(a) Valid tree

(b) Tree to list


Figure 9 Valid tree and Tree to list

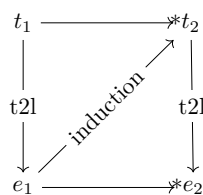
substitution of the alternative we are building and is updated when going to the rhs of the *Or* constructor. The *bt* list represents the future alternatives list that have the same depth of the current tree root. They are useful to know if they should be added to the current goal as cut alternatives.

An *OK* node represent a success in a tree, then the translation of this tree returns a singleton list with the couple $(s, [::])$: there is one alternative with 0 goals, i.e. a success in elpi by STOPE. The *KO* node represent a failure, and, therefore 0 alternatives. An atom is translated into a singleton with the atom as first goal and the empty cut-alternative list: we are not adding *bt* since they are the alternatives for the right-sibling, not the right-uncles, this means that if we have *a* is a cut, then it erases the alternatives *bt*.

In a disjunction, we are scendendo di un livello the distance from our backtracking list. This means that *bt* becomes the list of right-uncles of the two branches of the *Or* constructor. The compilation of the rhs is done independently of if the lhs exists: we transform the rhs into an elpi state and passing the empty list of alternatives, since the rhs as no right-siblings. Then, thanks to `add_ca_deep`, we recursively add *bt* to every cut-alternative appearing in the translated goals. If the lhs exists, we translate it and pass the translation of the rhs as the list of right-siblings. Then we prepend the translated lhs to the translated rhs and add *bt* with `add_ca_deep`.

We compile the *And* case by translating its lhs to the list *lA* and reset point to the list *lB0*. Then, thanks to `add_deep`, we recursively append *lB0* to the alternatives born in *lA*, i.e. we leave unchanged the pointers to *bt*, if any, in *lA*. We finally need to treat cases whenever the list *lA* is empty or not. In the former case the empty list is returned, otherwise, we have a list $[:: (s_0, gs) \& al]$. By the semantics of the *And* in the *runT* interpreter, the rhs of the *And* represent the sequent of goals to be executed after the first alternative in the lhs, it corresponds to the list of goals *gs*. The reset point *B0* is to be appended to the next

alternative in
tree? rephrase



■ **Figure 10** Induction scheme for Lemma 10

alternatives in the lhs, that is the queue of lA , that is al .

We note therefore that the compilation of lB is done by passing the alternatives $al + bt$ since the alternatives in al are to be...

5.3 Equivalence theorems

We have now all the ingredients that ...

```

Lemma tree_to_elpi fv A s1 b fv' r :
  vars_tree A `<=` fv → vars_sigma s1 `<=` fv →
  valid_tree A →
  runT u p fv s1 A r b fv' →
  let xs := t2l A s1 [::] in
  let r' := omap (fun '(s, t) => (s, (t2l (odflt KO t) s1 [::]))) r in
  runE u p fv xs r'.

```

(9)

```

Lemma elpi_to_tree v0 a r :
  runE u p v0 a r →
  ∀ s0 t, valid_tree t → t2l t s0 [::] = a →
  ∃ b v1,
  if r is Some (s1, a') then
    ∃ t1, runT u p v0 s0 t (Some (s1, t1)) b v1 ∧ t2l (odflt KO t1) s0 [::] = a'
  else runT u p v0 s0 t □ b v1.

```

(10)

The proof of Lemma 10 is based on the idea explained in [2, Section 3.3]. An ideal statement for this lemma would be: given a function $l2t$ transforming an elpi state to a tree, we would have have that the the execution of an elpi state e is the same as executing $runT$ on the tree resulting from $l2t(e)$. However, it is difficult to retrieve the strucutre of an elpi state and create a tree from it. This is because, in an elpi state, we have no clear information about the scope of an atom inside the list and, therefore, no evident clue about where this atom should be place in the tree.

Our theorem states that, starting from a valid tree t which translates to a list of alternatives $(\sigma_1, g) :: a$. If we run in elpi the list of alternatives, then the execution of the tree t returns the same result as the execution in elpi. The proof is performed by induction on the derivations of the elpi execution. We have 4 derivations.

We have 4 case to analyse:

6 Case study: determinacy alanysis

we mechanize the first order part of xxx.

snippet det, main thm, invariant det tree (valid tree prev section?)

proof induciton on exec, step/next alt preserving invariant proved by induction on the tree.

with list semantics cut and next alt requires to express a link between the ca or next alts and the current goal, which is non trivial without an intermediate data strature like the tree

7 Related work

The only mechanization of Prolog's semantics we could find is the one by Pusch in ISABELLE/HOL [16] that is based on the model of Debray and Mishra [6, Sec. 4.3]. Their work start from that semantics and refines it by introducing some optimizations usually found in the Warren abstract machine [20, 1]. They do not use the semantics to prove properties on the execution of programs as we do in our use case, hence they do not face the difficulty described in seciton XXX that pushed us to develop an more explicit semantic (with respect to cut). In some way our work is complementary, showing he stack based semantic equivalent to more optimized one, we could consider implement in Elpi.

Another relevant work are the ones by King cite, but we could not find their code. Their semantics is denotational and cannot easily cover all the features our paper XX does, but would have been sufficient for this first step.

The proof technique we use to relate the two semantics is inspired by XXX and we thank Y.Bertot for insightful discussions on the topic. In XXX Bertot relates bla bla, witout never constructing a decompiler, exactly as we never build a list to tree function.

8 Conclusion

We do this. This is a first step for XXXX. We need to add substitutions. The missing part of the proof becomes related to abstract interpretation, since the det types for a lattice and one has to prove that the concrete runtime values variables take agree with the abstraction computed statically. This proof is ongoing and somehow orthogonal to the current one since the cut operator and the HO feature do not interfere with eachother in complex ways.

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